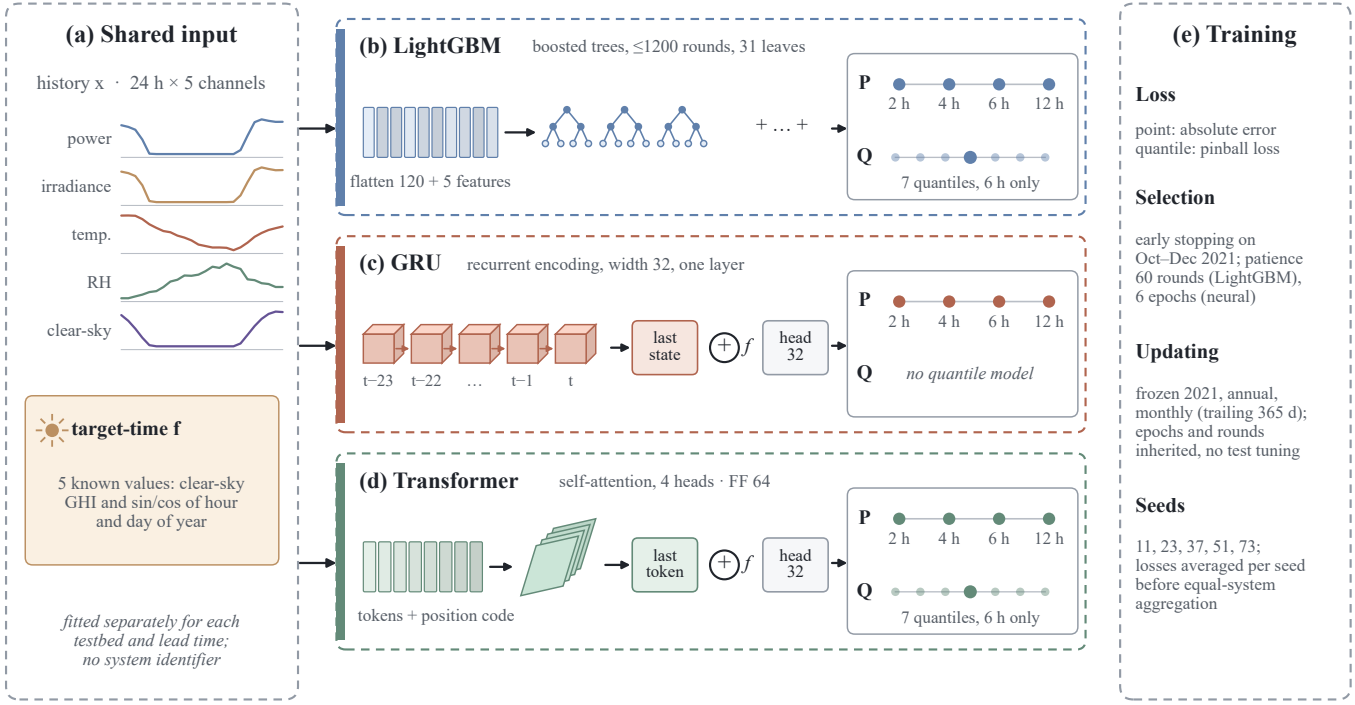

(a) Shared input
history x · 24 h × 5 channels
power
irradiance
temp.
RH
clear-sky
target-time f
5 known values: clear-sky GHI and sin/cos of hour and day of year
fitted separately for each testbed and lead time; no system identifier
(b) LightGBM
boosted trees, ≤1200 rounds, 31 leaves
flatten 120 + 5 features
+ … +
P
2 h
4 h
6 h
12 h
Q
7 quantiles, 6 h only
(c) GRU
recurrent encoding, width 32, one layer
t−23
t−22
…
t−1
t
last state
f
head 32
P
2 h
4 h
6 h
12 h
Q
no quantile model
(d) Transformer
self-attention, 4 heads · FF 64
tokens + position code
last token
f
head 32
P
2 h
4 h
6 h
12 h
Q
7 quantiles, 6 h only
(e) Training
Loss
point: absolute error
quantile: pinball loss
Selection
early stopping on Oct–Dec 2021; patience 60 rounds (LightGBM), 6 epochs (neural)
Updating
frozen 2021, annual, monthly (trailing 365 d); epochs and rounds inherited, no test tuning
Seeds
11, 23, 37, 51, 73; losses averaged per seed before equal-system aggregation